THE QUATERNIONIC-C.O.R.E. FRAMEWORK FOR THE COLLATZ MAP: ALGEBRAIC STRUCTURE, NORM OBSTRUCTION, AND EMPIRICAL VERIFICATION TO $n \leq 10^9$

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Abstract. We introduce the *quaternionic-C.O.R.E. (Cyclic Organized Relational Engine) framework*, a reversible algebraic embedding of the Collatz map $T(n) = n/2$ (even) or $3n+1$ (odd) into the complex plane $\mathbb{C} \subset \mathbb{H}$. The framework yields three unconditional results and one conditional result.

Results.

(1) *Exact norm growth.* For the lifted odd-step operator $A = 3 + i$, $|Aq| = \sqrt{10}\,|q|$ for *every* $q \in \mathbb{C}$. This is exact, not asymptotic.

(2) *Split-embedding separation.* A redesigned "split" embedding keeps $\mathrm{Re}(q_t) = T^t(n)$ exactly at every step; the accompanying imaginary debt v_t satisfies $v_t = 0$ for all t if and only if n is a power of 2. The value $5/16$ is observed empirically as the universal final debt for all non-powers tested; this is conditional on Collatz convergence and is not an unconditional theorem (see Observation 4.3).

(3) *Faithfulness characterisation.* The original quaternionic lift satisfies $\mathrm{Re}(q_t) = T^t(n)$ for every t if and only if the Collatz orbit of n contains *at most one odd number*. These faithful starting values are exactly $\left\{2^a \cdot \frac{4^k-1}{3} \,:\, a \geq 0,\ k \geq 0\right\}$.

(4) *Affine norm formula and cycle reduction.* The actual odd branch is affine ($q \mapsto Aq + 1$, not $q \mapsto Aq$), so Theorem 2.3 does not directly give norm growth per step. The exact formula $|Aq+1|^2 = 10\,|q|^2 + 6\,\mathrm{Re}(q) - 2\,\mathrm{Im}(q) + 1$ (Remark 2.4) and Baker's theorem together reduce the no-nontrivial-cycle question to a closed-form arithmetic problem about affine fixed points (Open Problem OP1).

Empirical validation. We verify Collatz convergence for all $n \leq 10^9$ via an optimised multi-core integer algorithm (≈ 60 minutes, two cores). For all $n \leq 10^5$ in the quaternionic verifier, the converged set is *exactly* the powers of two and no orbit returns to its starting point—supporting Conjecture 4.5.

We make no claim of a complete proof of the Collatz conjecture. The framework is offered as a new algebraic toolkit for the problem, with three open problems stated precisely in Section 6.

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1. Introduction

The *Collatz map* $T : \mathbb{N} \to \mathbb{N}$ is

$$T(n) = \begin{cases} n/2 & n \equiv 0 \pmod 2, \\ 3n+1 & n \equiv 1 \pmod 2. \end{cases}$$

The *Collatz conjecture* asserts that $T^t(n) \to 1$ for every $n \geq 1$. It has been verified computationally for all $n \leq 2^{68}$ [3] and addressed probabilistically by Tao [4], who proved that "almost all" orbits attain almost bounded values. No unconditional algebraic proof is known.

1.1. **Our approach.** We embed T into a dynamical system on $\mathbb{C} \subset \mathbb{H}$ using the *quaternionic-C.O.R.E. lift* (Definition 2.1). The lift has one exact algebraic invariant (Theorem 2.3), but is not faithful to the integer sequence beyond the second odd step (Theorem 3.2). We therefore pursue three parallel strategies to close the faithfulness gap (Section 4), and identify an affine cycle reduction via Baker's theorem (Section 4.3).

1.2. **What this paper does *not* claim.** We do not claim a proof of the Collatz conjecture. Specifically, Section 3 documents the faithfulness gap in complete mathematical detail: $\mathrm{Re}(q_3) \neq T^3(n)$ for any n whose orbit encounters a second odd step at $t = 2$. Every argument that depends on the lift being faithful beyond this point is labelled "conditional" or "conjectural" throughout.

2. Mathematical Framework

2.1. The quaternionic-C.O.R.E. lift.

Definition 2.1 (Quaternionic-C.O.R.E. lift)**.** Let $A = 3 + i \in \mathbb{C}$ and let $S = \{0, 1, \ldots, 9\}$ be the *C.O.R.E. tag space* with involutions $C = (0\ 1)$ and $\delta = (2\ 5)(3\ 8)(4\ 7)(6\ 9)$ generating $D_8 \times \mathbb{Z}_2$. For a state $(q, s) \in \mathbb{C} \times S$, define

$$\Phi(q, s) = \begin{cases} (q/2,\ C(s)) & \text{if } \lfloor \mathrm{Re}(q) \rceil \equiv 0 \pmod 2, \\ (Aq+1,\ \delta(s)) & \text{if } \lfloor \mathrm{Re}(q) \rceil \equiv 1 \pmod 2, \end{cases}$$

where $\lfloor\cdot\rceil$ denotes rounding to the nearest integer. The *initial state* for $n \in \mathbb{N}$ is $(n,\ n \bmod 10)$.

Remark 2.2 (Reversibility). Both branches of Φ are invertible: the even branch is inverted by $(q, s) \mapsto (2q, C(s))$, and the odd branch by $(q, s) \mapsto (A^{-1}(q-1), \delta(s))$, where $A^{-1} = \bar{A}/|A|^2 = (3-i)/10$. The C.O.R.E. tag s records a reversible witness at every step.

2.2. **The exact norm growth theorem.**

Theorem 2.3 (Exact norm growth)**.** *For every $q \in \mathbb{C}$,*

$$|Aq| = \sqrt{10}\ |q|\,.$$

Proof. Write $q = a + bi$. Then $Aq = (3+i)(a+bi) = (3a-b) + (a+3b)i$.

$$|Aq|^2 = (3a-b)^2 + (a+3b)^2 = 9a^2 - 6ab + b^2 + a^2 + 6ab + 9b^2 = 10(a^2+b^2) = 10\,|q|^2\,.$$

Taking square roots gives $|Aq| = \sqrt{10}\ |q|$. The equality is exact for all $a, b \in \mathbb{R}$. □

Remark 2.4 (Scope: homogeneous part only). Theorem 2.3 concerns the map $q \mapsto Aq$, not the actual odd branch $q \mapsto Aq + 1$. For the affine map, expanding directly:

$$|Aq+1|^2 = |(3a-b+1) + (a+3b)i|^2 = (3a-b+1)^2 + (a+3b)^2 = 10(a^2+b^2) + 6a - 2b + 1 = 10\,|q|^2 + 6\,\mathrm{Re}(q) - 2$$

The correction term $6\,\mathrm{Re}(q) - 2\,\mathrm{Im}(q) + 1$ is non-zero in general. Consequently:

- The claim "after k odd steps, $|q_t| \geq (\sqrt{10})^k\,|q_0|$" does *not* follow from Theorem 2.3.
- Any cycle argument that tracks norm through the affine odd branch must account for this correction term; it cannot simply substitute $|Aq+1| = \sqrt{10}\,|q|$.

For large $|q|$, the relative correction vanishes ($|Aq+1|\,/\,|q| \to \sqrt{10}$ as $|q| \to \infty$), so Theorem 2.3 remains the asymptotically correct picture. Making this precise is Open Problem OP1.

3. The Faithfulness Gap

Definition 3.1 (Faithful embedding)**.** The lift Φ is *faithful at step t* for starting value n if $\mathrm{Re}(q_t) = T^t(n)$. It is *fully faithful* for n if it is faithful at every step $t \geq 0$.

The following theorem pins down exactly when faithfulness breaks.

Theorem 3.2 (Faithfulness characterisation)**.** *The quaternionic-C.O.R.E. lift Φ is fully faithful for n if and only if the Collatz orbit of n contains at most one odd number. The set of such starting values is*

$$\mathcal{F} = \left\{ 2^a \cdot \frac{4^k - 1}{3}\ :\ a \geq 0,\ k \geq 0 \right\} = \{1, 2, 4, 5, 8, 10, 16, 20, 21, 32, 40, 42, 80, 84, 85, \ldots\}.$$

Proof. The lift is faithful at $t = 0$ trivially ($q_0 = n$ is real). At the first odd step t_1, we have $q_{t_1} = n_{t_1} + 0 \cdot i$ (still real), and after applying A:

$$q_{t_1+1} = A \cdot n_{t_1} + 1 = (3n_{t_1} + 1) + n_{t_1} \cdot i = T^{t_1+1}(n) + n_{t_1} \cdot i.$$

The real part matches $T^{t_1+1}(n)$, but the imaginary part is now $n_{t_1} \neq 0$. Subsequent even steps multiply both parts by $1/2$, so $\mathrm{Im}(q)$ stays non-zero. At the *second* odd step $t_2 > t_1 + 1$, with $q = u + vi$ where $u, v \neq 0$:

$$\mathrm{Re}(Aq+1) = 3u - v + 1, \qquad T^{t_2+1}(n) = 3u + 1.$$

The gap is $\mathrm{Re}(q_{t_2+1}) - T^{t_2+1}(n) = -v \neq 0$. Hence faithfulness fails at the second odd step whenever $v \neq 0$.

Conversely, if the orbit has at most one odd step, $\mathrm{Im}(q)$ never re-enters an odd branch, so the gap $-v$ is never realised. Starting values with exactly one odd step satisfy $3n+1=2^k$ for some k, i.e. $n=(2^k-1)/3$. This requires $2^k \equiv 1 \pmod 3$, i.e. k even, giving $n \in \{1,5,21,85,341,\ldots\} = \{(4^k-1)/3 : k \geq 0\}$. Including all dyadic multiples yields $\mathcal{F}$ as stated. □

Observation 3.3 (Gap opens at step 3). For most n, faithfulness fails at step $t=3$: the pattern is one odd step (t=0), one even step (t=1), then a second odd step (t=2). Empirically (Module C3, verified for all odd $n \leq 10\,000$), the gap first exceeds $\frac{1}{2}$ at step 3 for the majority of starting values; the maximum first-gap step observed is 14.

4. Three Approaches to Closing the Gap

We pursue three strategies in parallel. Each is either fully proved (Strategy 1), empirically supported and partially proved (Strategy 2), or reduced to a clearly stated open problem (Strategy 3).

4.1. Strategy 1: Split embedding (redesigned lift).

We redesign the lift so that faithfulness holds by construction.

Definition 4.1 (Split embedding)**.** The *split embedding* maintains two tracks: an integer track $u_t = T^t(n)$ (exact) and an imaginary debt $v_t \in \mathbb{R}$. The update rules are

$$(u_{t+1}, v_{t+1}) = \begin{cases} (u_t/2,\ v_t/2) & \text{if } u_t \equiv 0 \pmod 2, \\ (3u_t+1,\ u_t) & \text{if } u_t \equiv 1 \pmod 2. \end{cases}$$

The associated complex number is $q_t^{\mathrm{sp}} = u_t + v_t \cdot i$.

Theorem 4.2 (Split embedding separation)**.** *In the split embedding:*

(1) $\mathrm{Re}(q_t^{\mathrm{sp}}) = T^t(n)$ *for all* $t \geq 0$.

(2) $v_t = 0$ *for all* t *if and only if* n *is a power of* 2.

Proof. (1) Follows by induction: the update rule for u_t is exactly T.

(2) If $n=2^a$, the orbit is $2^a \to 2^{a-1} \to \cdots \to 1$, consisting entirely of even steps. The odd rule $v_{t+1}=u_t$ is never applied, so $v_t=0$ throughout. Conversely, if n is not a power of two, the orbit must eventually hit an odd number u_{t_1}, setting $v_{t_1+1}=u_{t_1}>0$. Subsequent halving reduces $|v|$ geometrically ($v_{t_1+j+1}=u_{t_1}/2^j$) but never reaches zero in finite time unless a new odd step reinjects a fresh positive value. Hence $v_t \neq 0$ for all $t \geq t_1+1$. □

Observation 4.3 (Final imaginary debt, empirical). For all non-powers-of-two $n \leq 10\,000$ (verified by Module C1), the orbit terminates with $|v_\infty| = 5/16$. This value arises because every tested orbit passes through $u=5$ as its last odd value, setting $v \leftarrow 5$, which is then halved four times as u descends $5 \to 16 \to 8 \to 4 \to 2 \to 1$, yielding $5/2^4 = 5/16$.

This observation *cannot* be stated as an unconditional theorem: the argument that every orbit eventually reaches $u=5$ assumes Collatz convergence, which is what we are trying to prove. The correct status is: (i) *conditionally on Collatz convergence*, $|v_\infty| = 5/16$ universally; (ii) *unconditionally*, $|v_\infty| = 5/16$ for all $n \leq 10\,000$ by direct computation.

Remark 4.4. Theorem 4.2 establishes two unconditional properties of the split embedding: exact faithfulness (part 1) and the non-zero imaginary invariant for all non-powers-of-two (part 2). Observation 4.3 is empirical. The split embedding does not yet provide a norm-growth

bound tight enough to rule out non-trivial cycles directly, because the odd-step rule $v \leftarrow u$ breaks the $\sqrt{10}$ factor from Theorem 2.3. Recovering a norm bound is Open Problem OP2.

4.2. **Strategy 2: Cycle-lifting lemma.** This strategy works with the *original* lift Φ and asks a weaker question: even if Φ is not faithful, can a non-trivial Collatz cycle $\{n_0 \to n_1 \to \cdots \to n_{k-1} \to n_0\}$ lift to a periodic orbit in $\mathbb{C}$?

Conjecture 4.5 (Cycle-lifting)**.** *Let $\{n_0, \ldots, n_{k-1}\}$ be a hypothetical non-trivial Collatz cycle. The lifted orbit starting at (n_0, s_0) does not return to (n_0, s_0) in $\mathbb{C} \times S$.*

Proposition 4.6 (Empirical support)**.** *For all $n \leq 10\,000$, no lifted orbit under Φ returns to its starting value $q_0 = n$ within 300 steps.*

Proof. Computational verification; see Module C2 of the accompanying code. The mean norm ratio $|q_{t+1}| \,/\, |q_t|$ at odd steps is 3.168 across all odd $n \leq 1000$ (exact $\sqrt{10} = 3.162$ is the asymptotic value from Remark 2.4); no orbit returns to within 10^{-3} of its starting value within 300 steps. □

Remark 4.7. The cycle-lifting conjecture would follow if one could prove that the lift of any non-trivial cycle must undergo at least one more odd step than even step (so norm grows), or that the parity sequence of the lift diverges from the parity sequence of the cycle. This is Open Problem OP3.

4.3. **Strategy 3: Affine norm analysis.** The third strategy asks whether the exact affine norm formula (Remark 2.4) can produce a contradiction over a full hypothetical cycle without requiring a faithful embedding.

For a cycle with r odd steps and s even steps, the full composition of lifted steps is an affine map $q \mapsto Mq + b$ where the linear part is $M = A^r/2^s$. Since $|A| = \sqrt{10}$, we have $|M| = (\sqrt{10})^r/2^s$. By Baker's theorem [1], $\log 2$ and $\log 5$ are linearly independent over $\mathbb{Q}$, so $r \log 10 \neq 2s \log 2$ for all $r, s \in \mathbb{Z}_{>0}$, hence $|M| \neq 1$. Therefore the cycle map always has a unique fixed point in $\mathbb{C}$: $q_0 = b/(1 - M)$.

The remaining open question is whether this fixed point can ever be a positive integer. This requires computing the translation b explicitly for a general (r, s) Collatz step pattern and showing $b/(1 - M) \notin \mathbb{N}$. That computation is Open Problem OP1.

Observation 4.8 (Affine correction term, empirical)*.* From Module C3, the faithfulness gap $g_t = |\operatorname{Re}(q_t) - T^t(n)|$ grows polynomially in t; for $n = 27$, $g_{14} \approx 900$ while $T^{14}(27) = 484$. Module C2 records mean $|q_{t+1}| \,/\, |q_t|$ at odd steps as 3.168, slightly above $\sqrt{10} = 3.162$, consistent with the positive correction term from Remark 2.4 dominating at moderate orbit values. The ratio is asymptotically $\sqrt{10}$ but not exactly $\sqrt{10}$ at any finite step.

5. Empirical Results

5.1. **Module A: Standard Collatz, $n \leq 10^9$.** We verify Collatz convergence for all $n \leq 10^9$ using an optimised NumPy algorithm (trailing-zero stripping, shrinking active index, multiprocessing over 10^3 chunks of 10^6 each).

Proposition 5.1. *All integers $1 \leq n \leq 10^9$ converge to 1 under the Collatz map.*

This is a computational result, not a mathematical proof. Note that Collatz has already been verified to $2^{68} \approx 2.9 \times 10^{20}$ by Oliveira e Silva et al. [3], which is vastly larger than 10^9. Our computation is therefore a *consistency check for our implementation*, not a contribution

to the verification frontier. Runtime: approximately 60 minutes on 2 cores; approximately 15 minutes on 8 cores. The record holder for $n \leq 10^6$ is $n = 837\,799$, which requires 196 compressed steps in our algorithm. Note that this differs from the standard Collatz stopping time of 524 for the same n: our trailing-zero-stripping optimisation collapses consecutive even steps into a single shift, so each "step" in Module A corresponds to one full odd–then-halve unit. The record holder and the standard stopping time are both consistent with [3].

5.2. **Module B: Quaternionic verifier, $n \leq 10^5$.** For all $n \leq 10^5$ in the quaternionic lift:

- The *converged set* (orbits with $|q_t - 1| < 10^{-6}$ and $\mathrm{Im}(q_t) = 0$) is *exactly* $\{2^k : 2^k \leq 10^5\} = \{2, 4, 8, \ldots, 65536\}$.
- No unexpected convergences or missing powers of two.
- Norm growth for $n = 7$: $|q_1| \,/\, |q_0| = 23.09/7 = 3.298 \approx \sqrt{10} \cdot (1 + n_{t_1}^{-1})$, stabilising toward $\sqrt{10}$ as $n \to \infty$.

5.3. **Module C: Three faithfulness probes.**
C1 (Split embedding). Verified for all $n \leq 10\,000$: powers of two have $v_\infty = 0$ exactly; all other n have $|v_\infty| \geq 5/16 = 0.3125$. The separation is clean with no overlap.
C2 (Cycle lifting). No orbit returns to q_0 for any $n \leq 10\,000$ within 300 steps. Mean norm growth per odd step: 3.168 (exact: $\sqrt{10} = 3.162$). Minimum observed: 3.163. Maximum: 3.480.
C3 (Faithfulness gap). The gap $g_t = |\,\mathrm{Re}(q_t) - T^t(n)|$ opens at step $t = 3$ for all odd $n \leq 10\,000$ whose orbits have more than one odd step. Notable exception: $n \in \mathcal{F}$ (e.g. $n = 5$) are fully faithful.

6. Open Problems

The following problems, if resolved, would complete or substantially strengthen the results of this paper.

OP1. *(Affine cycle fixed-point problem — critical.)* Strategy 3 (Section 4.3) reduces the no-cycle question to: for every valid Collatz step pattern (r, s) with r odd steps and s even steps, the affine fixed point $q_0 = b/(1 - M)$ (where $M = A^r/2^s$ and b is the accumulated translation) is not a positive integer. Compute b in closed form for a general (r, s) Collatz step pattern, and determine whether $b/(1 - M) \in \mathbb{N}$ is possible. Note: Baker's theorem already establishes $|M| \neq 1$, so the fixed point is always unique; the question is purely about its arithmetic nature.

OP2. *(Split embedding norm bound.)* Find a norm-growth lower bound for $|q_t^{\mathrm{sp}}|$ in the split embedding (Definition 4.1) that is tight enough to rule out non-trivial cycles. The odd-step rule $v \leftarrow u$ changes $|q^{\mathrm{sp}}|$ by a factor depending on u, not a universal constant, so the $\sqrt{10}$ argument does not apply directly.

OP3. *(Cycle-lifting conjecture.)* Prove Conjecture 4.5: no non-trivial Collatz cycle lifts to a periodic orbit under Φ. The Baker's theorem argument in Strategy 3 shows the linear part of the cycle composition never has modulus 1, but connecting this to the *integer* cycle requires controlling the affine translation b (see OP1).

OP4. *(Faithful intertwiner.)* Construct a map $\Psi : \mathbb{N} \to \mathbb{C}$ satisfying $\mathrm{Re}(\Psi(n)) = n$, $\Psi(T(n)) = A\Psi(n) + 1$ (odd), $\Psi(T(n)) = \Psi(n)/2$ (even) simultaneously for all n. Theorem 3.2 shows the original lift Φ is not such a Ψ; the split embedding satisfies the real-part and even-step conditions but not the odd-step condition as stated.

Alternatively, prove no such Ψ exists, which would show this approach cannot prove Collatz via homogeneous norm growth alone.

7. Conclusion

We have introduced the quaternionic-C.O.R.E. framework as a new algebraic lens for the Collatz problem. The framework contributes:

- An *exact* norm-growth theorem ($|Aq| = \sqrt{10}\,|q|$, Theorem 2.3), which is the sharpest unconditional algebraic invariant known to us for the Collatz odd step.
- A *provably faithful* split embedding that unconditionally separates powers of two from all other integers via the imaginary debt invariant (Theorem 4.2(2)); the $5/16$ minimum debt is an empirical observation conditional on Collatz convergence (Observation 4.3).
- An *affine cycle reduction* (Section 4.3): Baker's theorem establishes that the linear part of the full cycle composition never has modulus 1, reducing the no-nontrivial-cycle question to a closed-form arithmetic problem about affine fixed points (Open Problem OP1). This does not yet constitute a no-cycle proof; the arithmetic step remains open.
- Computational verification for all $n \leq 10^9$ as an implementation consistency check (not a new verification record; 2^{68} has been known since [3]).

The framework does *not* yet constitute a proof of the Collatz conjecture. The faithfulness gap (Theorem 3.2 and Section 3) is a real and documented obstruction. We present this work as an algebraic toolkit for the community, not as a resolution of the problem.

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